

Solution Requirements

Date	29 September 2026
Team ID	SWTID-2026-1307
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users must log in to the ServiceNow instance with valid credentials (local login or SSO). Only authenticated users can reach the target table; unauthenticated requests are rejected before any ACL is evaluated.	High
2	Authorization levels	Roles: Admin / security_admin (create and maintain ACLs), Restricted Role Users (view records only when the field value matches their permission), Standard Users (blocked from records with the restricted field value), Auditor (read-only view of logs).	High
3	External interfaces	No third-party APIs required. Works within the ServiceNow platform (tables, ACL engine, Script Includes). Optional: LDAP / SSO for user identity and REST API / list / form access, which must respect the same ACL.	Low
4	Transactions processing	On every read / write / list request the ACL engine runs the script, checks the record's field value (e.g., current.priority, current.confidential) against the user's role and sets answer = true or false. Denied records are hidden from forms, lists, reports and APIs.	High

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
5	Reporting	Reports / dashboards on ACL access denials, records restricted by field value, and user access activity. Data export (CSV / PDF) must also exclude records the user cannot access.	Medium
6	Business rules, etc.	Records whose field has the defined value (e.g., Confidential = true) are visible only to authorised roles or the record owner. Script logic must be reusable through Script Includes, tested through Impersonate User, and moved between instances by update sets.	High
7	Compliance to laws or regulations	Follow data-privacy principles (GDPR, least privilege, need-to-know) and organisation security policy. Access decisions must be traceable for audit purposes.	Medium
8	Other	Provide a fallback deny-by-default ACL, clear error handling for scripts, and documented ACL scripts with comments for maintenance.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	ACL script must be lightweight (no heavy GlideRecord queries) so record and list loading is not slowed.	ACL evaluation adds < 200 ms; list loads in < 2 seconds
2	Scalability	ACL logic must work as record volume and users grow, using indexed fields and role checks instead of per-record loops.	Supports 10,000+ records and 1,000+ concurrent users
3	Security & Data Privacy	Restricted records must never be exposed through forms, lists, reports, search or REST APIs. Follow deny-by-default and least privilege; script must avoid client-controlled inputs.	0 unauthorized access in role-based tests; 100% restricted records hidden
4	Reliability & Availability	ACL must behave consistently on every request and fail safe (deny access) when the script errors.	99.9% uptime per month; no false grants on script error
5	Usability & Accessibility	Users without access should see a clear "Access denied / record not available" behaviour and admins get a simple ACL configuration form.	Complies with WCAG 2.1 AA; ACL created in < 15 minutes
6	Other	Maintainability: commented, reusable scripts with version control through update sets. Portability: ACL moves between Dev, Test and Prod without change.	100% ACLs documented; deployed with no manual re-work